

Task 1: DNA/Protein Sequence Analysis Using BLASTP

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DOMAIN: BIOINFORMATICS

PROTEIN SELECTED: HUMAN INSULIN (INS)

1. AIM:

To perform DNA/protein sequence analysis using BLAST and identify homologous sequences in biological databases.

2. TOOLS USED:

- UNIPROT DATABASE
- NCBI BLAST
- INTERNET BROWSER
- FASTA PROTEIN SEQUENCE

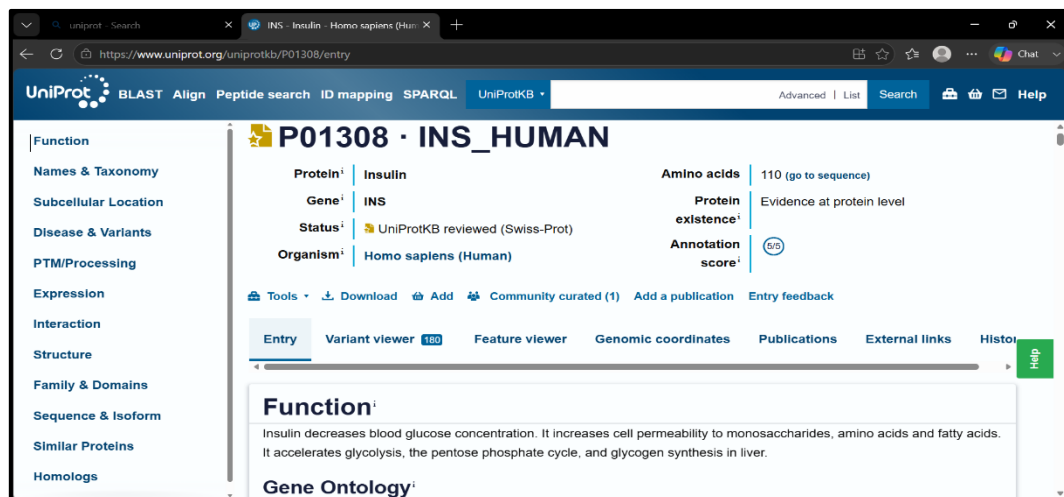


FIG1: Human Insulin protein entry retrieved from UniProt database.

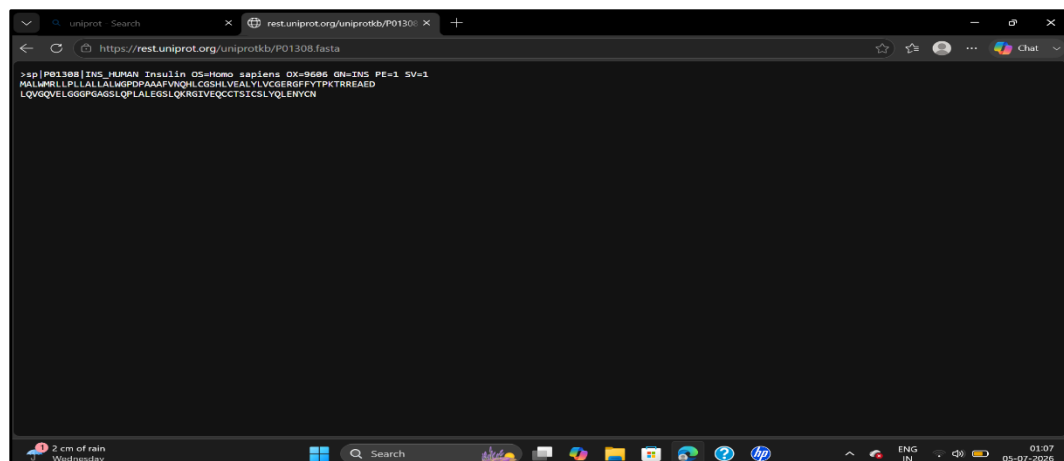


FIG2: FASTA format of Human Insulin protein sequence downloaded from UniProt.

3. Procedure: The Human Insulin (INS) protein sequence was downloaded from the *UniProt* database in FASTA format. The FASTA sequence was copied and submitted to the NCBI BLASTP tool. The default BLASTP parameters were used with the Non-Redundant (nr) protein database. The search was executed to identify homologous protein sequences from different organisms. The BLAST results were analyzed based on sequence identity, query coverage, E-value, and alignment score.

4. BLAST Search Steps

- Open the *NCBI BLAST* website.
- Select Protein BLAST (BLASTP).
- Paste the Human Insulin FASTA sequence into the query box.
- Keep the default database (nr) and default parameters.
- Click BLAST to start the search.
- Wait for the results to load.
- Analyse the homologous protein sequences.

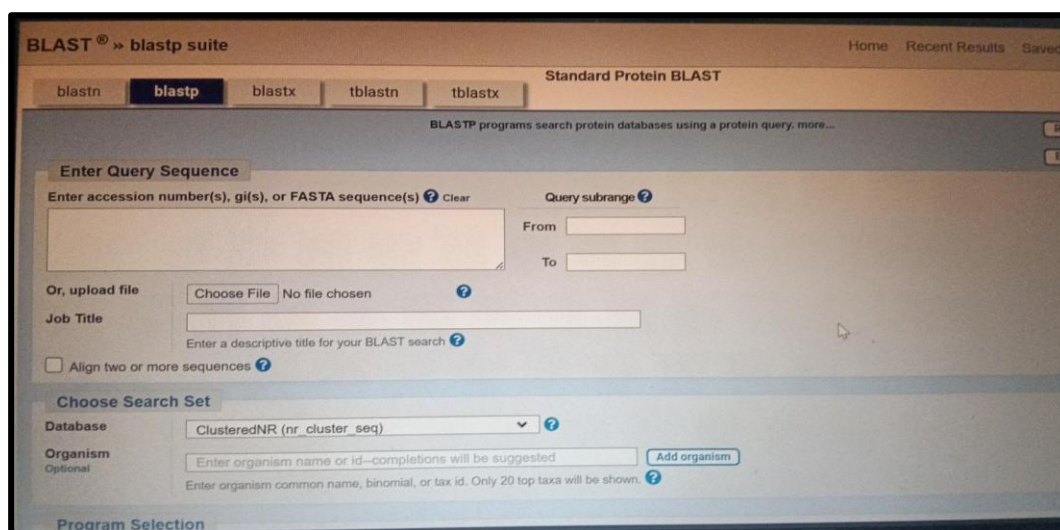
The image shows the NCBI BLASTP search interface. At the top, there's a navigation bar with 'BLAST' and 'blastp suite'. Below this, there are tabs for 'blastn', 'blastp', 'blastx', 'tblastn', and 'tblastx', with 'blastp' selected. The main section is titled 'Standard Protein BLAST'. It contains a 'Query Sequence' input field with a 'Clear' button and a 'Query subrange' section with 'From' and 'To' fields. Below the query field is an 'Or, upload file' section with a 'Choose File' button and a 'No file chosen' message. There's also a 'Job Title' input field and a checkbox for 'Align two or more sequences'. The 'Choose Search Set' section includes a 'Database' dropdown menu set to 'ClustredNR (nr_cluster_seq)' and an 'Organism' input field with an 'Add organism' button. At the bottom, there's a 'Program Selection' section.

FIG3: BLASTP SEARCH INTERFACE

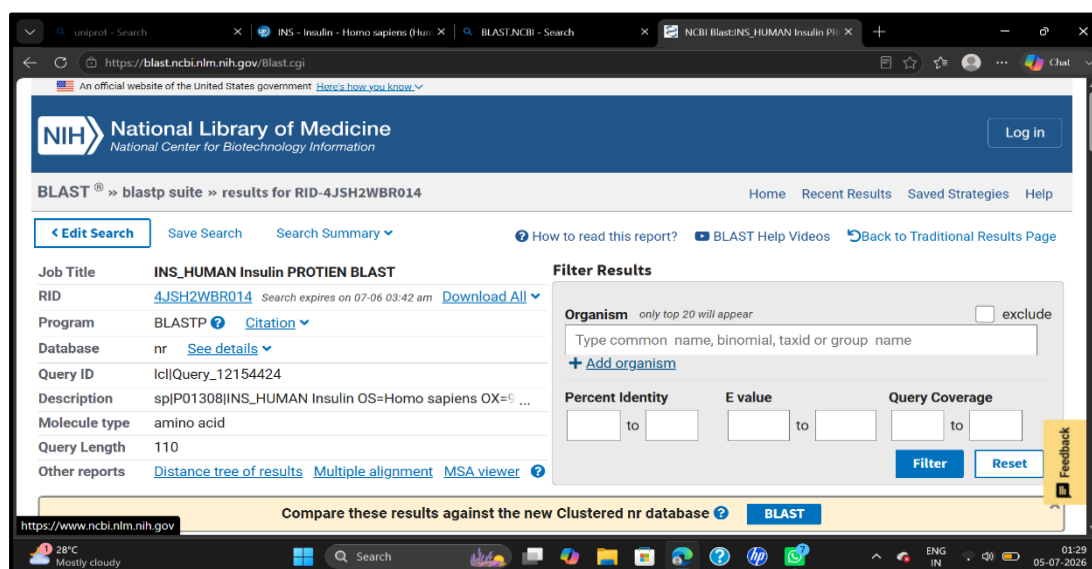
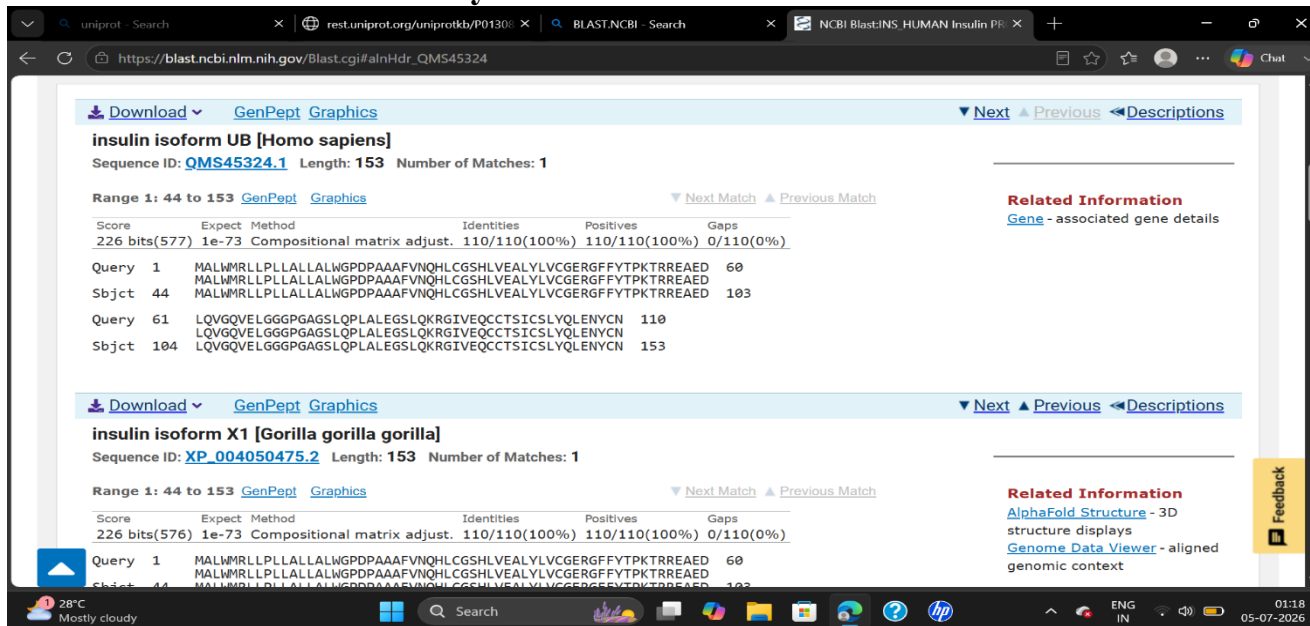
The image shows the NCBI BLASTP search results page. At the top, there's a navigation bar with 'BLAST' and 'blastp suite'. Below this, there are tabs for 'blastn', 'blastp', 'blastx', 'tblastn', and 'tblastx', with 'blastp' selected. The main section is titled 'Standard Protein BLAST'. It contains a 'Query Sequence' input field with a 'Clear' button and a 'Query subrange' section with 'From' and 'To' fields. Below the query field is an 'Or, upload file' section with a 'Choose File' button and a 'No file chosen' message. There's also a 'Job Title' input field and a checkbox for 'Align two or more sequences'. The 'Choose Search Set' section includes a 'Database' dropdown menu set to 'ClustredNR (nr_cluster_seq)' and an 'Organism' input field with an 'Add organism' button. At the bottom, there's a 'Program Selection' section.

FIG4: BLASTP search submitted successfully and processing on the NCBI server

5. BLASTP Results and Analysis:



PARAMETER	OBSERVATION
Protein selected	Human insulin
Query length	110 amino acids
Database Used	Non-Redundant (nr)
Top Homologous Sequence	Insulin isoform UB (Homo sapiens)
Percent Identity	100% (110/110)
Query Coverage	100%
E-value	1e-73
Alignment Score	226 bits
Similarity(positives)	100%

FIG6: BLASTP results showing the sequence alignment and search details for the HUMAN INSULIN PROTEIN

6. Analysis

The BLASTP analysis identified Human Insulin as the top matching protein with 100% identity, 100% query coverage, an E-value of 1e-73, and an alignment score of 226 bits. These results indicate a highly significant match and show that the insulin protein is highly conserved.

7. Conclusion

The Human Insulin protein sequence was successfully analysed using NCBI BLASTP. The results showed a perfect sequence match with high identity and query coverage, demonstrating the effectiveness of BLAST in identifying homologous protein sequences.

8. References

- UniProt Consortium. UniProt Knowledgebase (UniProtKB). <https://www.uniprot.org>
- National Center for Biotechnology Information (NCBI). Basic Local Alignment Search Tool (BLAST). <https://blast.ncbi.nlm.nih.gov>